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Lab sheet 1

JavaScript Fundamentals –

Variables, Operators, Control Structures, and Data Types

Course Outcome: *CO1: Use markup and scripting languages to design and validate dynamic web pages.*

Aim

To understand and apply basic JavaScript concepts such as variables, console output, type coercion, objects, arrays, operators, conditional statements, and looping statements

Question 1 – Variables, Console.log, and Coercion

Problem Statement
Write a JavaScript program that demonstrates variable declaration and type coercion.

Task
Declare two variables, one number and one string
Add them and display the result using console.log
Explain the output using comments

Sample Input:

let a = 10;

let b = "20";

Code:

```
JS assignment1_057.js > ...
let a = 10;
let b = "20";

let result = a + b;
console.log(result);

```

Output:

```
nandita-rajesh@nandita-rajesh-HP-Laptop-15s-eq2xxx:~/Desktop/S6/FULL-STACK
/assignment1$ node assignment1_057.js
1020
```

Explanation: In JavaScript, when the + operator is used with a number and a string, type coercion occurs.

Question 2 – Operators and Conditional Statements

Problem Statement
Write a JavaScript program to check whether a number is positive, negative, or zero.

Task
Use relational and logical operators
Use if–else conditional statements
Display the result using console.log

Sample Input

let num = -5;

Code:

```
let num = -5;

if (num > 0) {
  console.log("Positive number");
} else if (num < 0) {
  console.log("Negative number");
} else {
  console.log("Zero");
}
```

Output:

```
nandita-rajesh@nandita-rajesh-HP-Laptop-15s-eq2xxx:~/Desktop/S6/FULL-STACK
/assignment1$ node assignment1_057.js
Negative number
```

Explanation: The relational operators are used to compare the value of num. Since num is -5, the condition num < 0 is true. Hence, the program prints "Negative number".

Question 3 – Objects in JavaScript

Problem Statement
Create a JavaScript object to store student details and display specific properties.

Task
Create an object with name, rollNo, and marks
Print the student name and marks using console.log

Sample Code:

```
const student = {

  name: "Anil",

  rollNo: 23,

  marks: 85};
```

Code:

```
const student = {
  name: "Anil",
  rollNo: 23,
  marks: 85
};

console.log("Student Name:", student.name);
console.log("Marks:", student.marks);
```

Output:

```
nandita-rajesh@nandita-rajesh-HP-Laptop-15s-eq2xxx:~/Desktop/S6/FULL-STACK
/assignment1$ node assignment1_057.js
Student Name: Anil
Marks: 85
```

Explanation: An object named student is created with properties name, rollNo, and marks. Dot notation is used to access and print the name and marks properties.

Question 4 – Arrays and Looping Statements

Problem Statement
Write a JavaScript program to print all elements of an array using a loop.

Task
Create an array of five numbers
Use a for loop to print each element

Sample Code

let arr = [10, 20, 30, 40, 50];

Code:

```
let arr = [10, 20, 30, 40, 50];

for (let i = 0; i < arr.length; i++) {
  console.log(arr[i]);
}
```

Output:

```
nandita-rajesh@nandita-rajesh-HP-Laptop-15s-eq2xxx:~/Desktop/S6/FULL-STACK
/assignment1$ node assignment1_057.js
10
20
30
40
50
```

Explanation: An array of five numbers is created and a for loop is used to iterate through the array. Each element is printed.

Question 5 – Combined Concepts (Array + Operators + Condition + Loop)

Problem Statement
Write a JavaScript program to count how many numbers in an array are even and odd.

Task
Use modulo operator
Use if–else condition
Use a loop to traverse the array

Sample Code

let numbers = [12, 7, 9, 20, 18];

Code:

```
let numbers = [12, 7, 9, 20, 18];
let evenCount = 0;
let oddCount = 0;

for (let i = 0; i < numbers.length; i++) {
  if (numbers[i] % 2 === 0) {
    evenCount++;
  } else {
    oddCount++;
  }
}

console.log("Even numbers:", evenCount);
console.log("Odd numbers:", oddCount);
```

Output:

```
nandita-rajesh@nandita-rajesh-HP-Laptop-15s-eq2xxx:~/Desktop/S6/FULL-STACK
/assignment1$ node assignment1_057.js
Even numbers: 3
Odd numbers: 2
```

Explanation: A loop is used to traverse the array. The modulo operator (%) checks whether a number is even or odd. If a number is divisible by 2, evenCount is incremented else oddCount is incremented.